

Problem Statement

Objective:

Estimate **Heart Rate (HR)**, **Systolic Blood Pressure (SBP)**, and **Diastolic Blood Pressure (DBP)** from smartphone-acquired signals stored in .txt files, using corresponding labels provided in a master .csv file.

Dataset Description

- **Link:** [Click here for Dataset.](#)
 - **Signal Files:** Each subject's raw signal data is stored in individual .txt files.
 - **Master Data CSV:** A master_data.csv file containing the mapping of each subject's file name with their corresponding **HR**, **SBP**, and **DBP** values.
-

Tasks to Perform

1 Load the Data:

2 Preprocess the Signals:

3 Normalize the Signals:

4 Extract Features:

5 Perform Feature Selection:

6 Train a Machine Learning Model:

7 Evaluate the Model:

8 Test with Unseen Data:
